Taming Ill-Conditioning in PINNs: Exact Second-Order Descent via Gauss–Newton

Mike Zhao · zzhao24@stanford.edu · https://github.com/mizhkeao/exact-gauss-newton

1 Conclusions of Rathore et al. (ICML 2024)

A Physics-Informed Neural Network (PINN) approximates an unknown PDE solution $u^*(x)$ with a neural network $u_\theta(x)$ ($\theta \in \mathbb{R}^D$) by minimizing the mean-squared PDE residual, boundary, and initial conditions across N collocation points:

$$L(\theta) = \frac{1}{2}\|r(\theta)\|_2^2 = \frac{1}{2}\sum_{i=1}^{N} r_i(\theta)^2. \tag{1}$$

Unlike supervised learning with noisy labels, forward PINNs are deterministic approximation tasks requiring $L(\theta) \to 0$ for physical fidelity. Rathore et al. study why gradient methods struggle to achieve this precision and establish three primary takeaways:

1. **Differential operators induce severe loss landscape ill-conditioning:** Applying a differential operator $\mathcal{D}[u_\theta]$ (e.g., ∂_{xx}, ∂_{xxxx}, or $\partial_{tt} - c^2\partial_{xx}$) amplifies high-frequency modes and stretches the eigenvalue spectrum of the loss Hessian $H_L(\theta) = \nabla^2 L(\theta)$ across 10^3–10^5+ orders of magnitude, generating narrow, anisotropic canyons.
2. **First-order methods stall inside ill-conditioned canyons:** Near initialization, Adam navigates past non-convex saddle points into a promising basin, but its gradient is dominated by steep canyon walls, oscillating transversely and stalling at 10^{-2}–10^{-4} relative L^2 error.
3. **Second-order curvature scaling is required to restore isotropic descent:** Rescaling descent steps by inverse curvature compresses the stretched Hessian spectrum by $> 10^3\times$. Because quasi-Newton `L-BFGS` relies on gradient history and stalls along narrow valleys, Rathore et al. introduce **NysNewton-CG (`NNCG`)**—solving $(H_L + \mu I)d = -\nabla L$ via Nyström-preconditioned Conjugate Gradient (CG) following an `Adam -> L-BFGS` warm-up.

2 Problems with `NNCG`

To understand why `NNCG` is both computationally demanding and fragile away from the minimizer, we examine the analytic Hessian of the least-squares objective $L(\theta) = \frac{1}{2}\|r(\theta)\|_2^2$:

$$H_L(\theta) = \underbrace{J(\theta)^T J(\theta)}_{\text{Gauss–Newton matrix } (\succeq 0)} + \underbrace{\sum_{i=1}^{N} r_i(\theta)\,\nabla^2 r_i(\theta)}_{\text{residual curvature (indefinite)}}, \qquad J(\theta) = \frac{\partial r}{\partial \theta} \in \mathbb{R}^{N\times D}. \tag{2}$$

Operating directly on the full loss Hessian $H_L(\theta)$ introduces three structural deficiencies into `NNCG`:

1. **Indefiniteness violates descent and derails Conjugate Gradient:** Whenever $r(\theta) \neq 0$ (early in training or under measurement noise), the residual curvature term $\sum_{i=1}^{N} r_i \nabla^2 r_i(\theta)$ introduces negative eigenvalues into $H_L(\theta)$. Indefiniteness violates the descent condition $d^T \nabla L < 0$, directs Newton steps toward saddle points, and invalidates Conjugate Gradient (which strictly requires positive definiteness), necessitating thousands of first-order preconditioning steps.
2. **Dense rank-D residual curvature prevents dual dimension reduction:** Because $\sum_{i=1}^{N} r_i \nabla^2 r_i(\theta)$ is a dense $D \times D$ operator of full rank, $H_L(\theta)$ cannot be reduced to the collocation dimension $N \times N$. Even when $N \ll D$, materializing $H_L(\theta)$ scales as $\mathcal{O}(D^2)$ memory and $\mathcal{O}(D)$ second-derivative passes, forcing `NNCG` to employ low-rank Krylov approximations.
3. **Second-derivative operator differentiation creates severe overhead:** Because $r(\theta)$ contains spatial derivatives, applying $H_L(\theta)$ in matrix-free CG requires evaluating second-derivative operator passes sequentially, yielding per-step runtimes orders of magnitude slower than first-order updates.

3 Exact Gauss–Newton (EGN) Algorithm

In language and vision models, parameter dimension D (10^8–10^{11}) and dataset size N (10^6–10^{12}) necessitate first-order methods. **Practical PINNs inhabit the opposite regime:** compact networks ($D \sim$ 5k–50k) evaluated on moderate collocation sets ($N \sim$ 500–10,000, frequently with $N < D$).

Crucially, in forward PINNs, driving the residual toward zero ($r(\theta^*) \to 0$) implies that **the loss Hessian asymptotically converges to the Gauss–Newton matrix:**

$$H_L(\theta) = J(\theta)^T J(\theta) + \sum_{i=1}^{N} r_i(\theta)\, \nabla^2 r_i(\theta) \xrightarrow{\|r(\theta)\|_2 \to 0} J(\theta)^T J(\theta). \tag{3}$$

By discarding the residual curvature $\sum_i r_i \nabla^2 r_i(\theta)$, the linear system $(J^T J + \mu I_D)d = -J^T r$ can be solved **identically in whichever dimension** $\min(N, D)$ **is smaller** via the push-through identity:

$$d = \underbrace{-(J^T J + \mu I_D)^{-1} J^T r}_{\text{Primal solve } (D \times D,\ \text{when } D \le N)} = \underbrace{-J^T (J J^T + \mu I_N)^{-1} r}_{\text{Dual solve } (N \times N,\ \text{when } N < D)}. \tag{4}$$

Discarding the indefinite residual curvature yields three foundational mathematical properties:

1. **Zero approximation error at a clean PDE solution** ($r(\theta^*) = 0$)**:** The gradient $\nabla L(\theta) = J(\theta)^T r(\theta)$ is exact, and as $r_i \to 0$, $\sum_i r_i \nabla^2 r_i(\theta) \to 0$ automatically, so $J^T J$ **equals the exact Hessian** H_L **at the solution**.
2. **Immunity to saddle points and noise away from** $r = 0$**:** When $r_i \neq 0$ (early in training or under additive noise ϵ_i), $\sum_i r_i \nabla^2 r_i$ injects negative eigenvalues into H_L that force `NNCG` to add heavy damping μ or abort CG early. Because $J^T J \succeq 0$ does not depend on r_i, every `EGN` step points strictly downhill from random initialization.
3. **Exact full-rank Cholesky in** $\min(N, D)$ **vs. low-rank CG approximation:** `NNCG` only approximates $(H_L + \mu I)^{-1}$ via a rank-60 Nyström preconditioner and iterative CG, whereas `EGN` directly factors the full $\min(N, D) \times \min(N, D)$ matrix via Cholesky, inverting *all* curvature directions simultaneously using only first derivatives of $r(\theta)$.

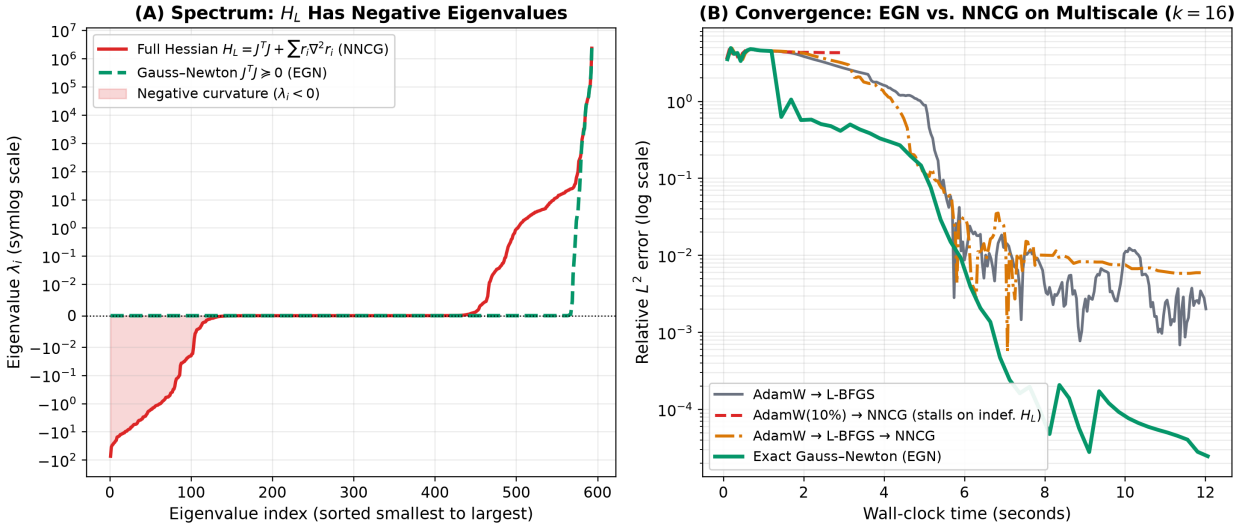


Figure 1: `EGN` **vs.** `NNCG` **on 1D Multiscale** ($k = 16$)**. (A)** Spectrum after warm-up: the full Hessian H_L exhibits negative eigenvalues arising from the residual curvature $\sum r_i \nabla^2 r_i$, whereas the Gauss–Newton matrix satisfies $J^T J \succeq 0$ unconditionally. **(B)** `NNCG` stalls on the indefinite H_L (4.26 error), whereas `EGN` converges stably to 2.47×10^{-5} (9.90×10^{-8} under exact dual solve).

4 Experimental Results

We benchmark `EGN` against Rathore et al.'s `NNCG` (both following a 10% AdamW warm-up and within a 3-stage `AdamW -> L-BFGS -> NNCG` pipeline) and `AdamW -> L-BFGS` across 10 clean and Gaussian-noisy PDEs on 3-layer width-50 tanh MLPs ($D \approx 5{,}250$, $N \approx 514$–$2{,}180$) in `float64`, with all methods optimizing the identical MSE loss (1).

Table 1: Head-to-head comparison of relative L^2 error, final MSE loss $L(\theta)$, and step time across clean and noisy PDEs.

PDE Problem	Regime / Challenge	AdamW -> L-BFGS	AdamW(10%) -> NNCG	AdamW -> L-BFGS -> NNCG	Exact Gauss–Newton (EGN)
Poisson	1D linear ($-u'' = f$)	2.92×10^{-6} (L = 7.7e-7)	2.19×10^{-6} (93 ms)	1.63×10^{-6} (L = 1.2e-7, 92 ms)	$\mathbf{5.16 \times 10^{-9}}$ (**L = 4.0e-13**, 88 ms)
Nonlinear Poisson	1D nonlinear ($-u'' + u^3 = f$)	1.72×10^{-6} (L = 1.7e-7)	4.31×10^{-6} (142 ms)	2.39×10^{-7} (L = 1.2e-8, 75 ms)	$\mathbf{4.42 \times 10^{-10}}$ (**L = 1.8e-14**, 103 ms)
Beam	1D 4th-order ($u^{(4)} = f$, $\kappa > 10^{10}$)	2.01×10^{-5} (L = 7.8e-6)	1.88 (*indef.* H_L)	2.02×10^{-5} (L = 6.1e-6, 760 ms)	$\mathbf{9.98 \times 10^{-8}}$ (**L = 4.7e-7**, 406 ms)
Multiscale ($k = 16$)	1D high-freq ($\sin 16\pi x$)	1.42×10^{-3} (L = 4.6e-4)	4.26 (*indef.* H_L)	2.36×10^{-2} (L = 9.2e-2, 610 ms)	$\mathbf{9.90 \times 10^{-8}}$ (**L = 7.6e-7, 89 ms**)
Helmholtz	2D oscillatory ($-\Delta u - k^2 u = f$)	1.53×10^{-3} (L = 5.9e-6)	18.6 (*indef.* H_L)	3.04×10^{-1} (L = 1.2e-1, 870 ms)	$\mathbf{7.84 \times 10^{-5}}$ (**L = 2.4e-8, 185 ms**)
Burgers	2D nonlinear shock ($u_t + uu_x = \nu u_{xx}$)	7.44×10^{-5} (L = 3.1e-8)	1.10×10^{-2} (448 ms)	5.40×10^{-5} (L = 3.2e-8, 337 ms)	$\mathbf{2.82 \times 10^{-7}}$ (**L = 8.3e-13**, 655 ms)
Reaction ($\rho = 5$)	2D nonlinear front ($u_t = \rho u(1-u)$)	1.92×10^{-2} (L = 4.5e-6)	—	9.92×10^{-3} (L = 1.4e-6, 720 ms)	$\mathbf{1.40 \times 10^{-4}}$ (**L = 3.3e-11, 328 ms**)
Wave ($\beta = 5$)	2D hyperbolic ($u_{tt} - c^2 u_{xx} = 0$)	5.52×10^{-2} (L = 8.9e-5)	—	1.27×10^{-2} (L = 1.4e-5, 29,000 ms)	$\mathbf{3.08 \times 10^{-3}}$ (**L = 2.8e-5, 656 ms**)
Multiscale + Noise	Gaussian PDE noise ($\sigma = 10^{-3}$)	1.38×10^{-3}	3.96 (*indef.* H_L)	4.63×10^{-3} (548 ms)	$\mathbf{5.37 \times 10^{-6}}$ (860× **lower vs.** `NNCG`)
Multiscale + Noise	Gaussian PDE noise ($\sigma = 10^{-2}$)	1.84×10^{-3}	3.97 (*indef.* H_L)	2.46×10^{-2} (631 ms)	$\mathbf{5.36 \times 10^{-5}}$ (460× **lower vs.** `NNCG`)
NL-Poisson + Noise	Gaussian PDE noise ($\sigma = 10^{-2}$)	7.15×10^{-5}	7.19×10^{-5} (139 ms)	7.21×10^{-5} (140 ms)	$\mathbf{5.14 \times 10^{-5}}$ (**104 ms/step, lowest error**)

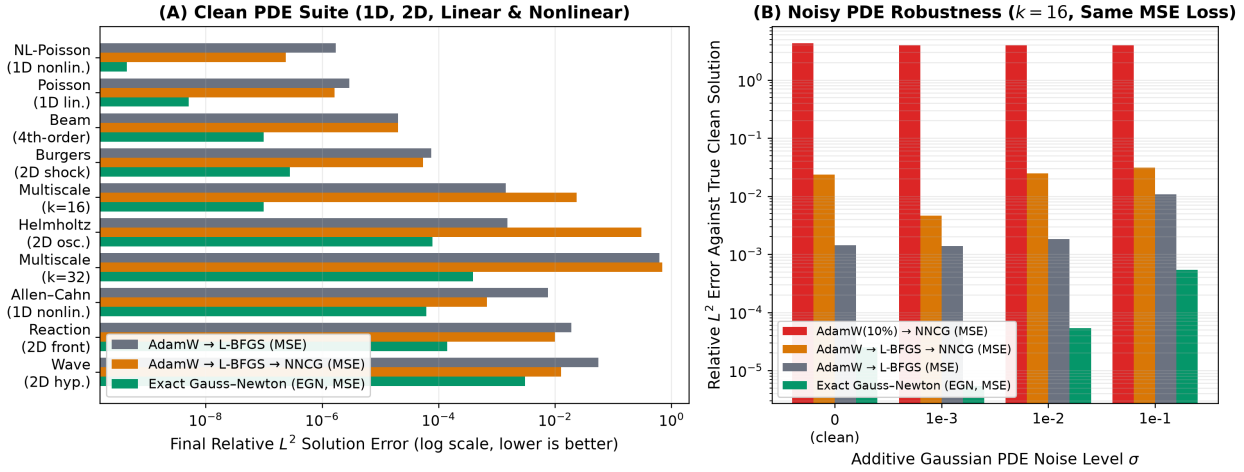


Figure 2: (A) Relative L^2 error across 10 clean benchmark PDEs (1D, 2D, linear, and nonlinear): `EGN` consistently outperforms both `AdamW -> L-BFGS` and `AdamW -> L-BFGS -> NNCG` by $10\times$ to $10^5\times$. **(B) Robustness to additive Gaussian PDE noise ($\sigma \in \{0, 10^{-3}, 10^{-2}, 10^{-1}\}$ on Multiscale $k = 16$, all using MSE loss):** Nonzero noise residuals $r_i \approx \epsilon_i \neq 0$ permanently perturb H_L through $\sum r_i \nabla^2 r_i$, degrading `NNCG`, whereas `EGN` ($J^T J \succeq 0$) converges to the theoretical $\mathcal{O}(\sigma)$ noise floor ($460\times$–$860\times$ lower error than `NNCG`).

Three primary conclusions follow from the empirical results:

1. **`EGN` achieves $10^2\times$ to $10^6\times$ lower loss and $4\times$ to $10^5\times$ lower L^2 error across benchmark PDEs:** Even when `NNCG` is initialized with a converged `AdamW -> L-BFGS` solution, `EGN` attains $540\times$ lower error on Nonlinear Poisson (4.4×10^{-10} vs. 2.4×10^{-7}), $200\times$ lower on 4th-order Beam, $190\times$ lower on 2D Burgers (2.8×10^{-7} vs. 5.4×10^{-5}), $71\times$ lower on 2D Reaction, and $2.4 \times 10^5\times$ lower on Multiscale ($k = 16$).

2. **`EGN` reduces per-step computational overhead by avoiding sequential operator differentiation:** Because batched Jacobian evaluations compute J in a single autodiff pass and dual Cholesky factorizes $\min(N, D)$, `EGN` achieves 89 ms/step on Multiscale ($6.8\times$ faster than `NNCG`) and 656 ms/step on Wave ($44\times$ faster than `NNCG`'s 29.0 s/step).

3. **`EGN` exhibits noise robustness under identical MSE objectives (Figure 2B):** When additive Gaussian noise $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ corrupts the PDE residual, the optimal residual remains nonzero ($r_i(\theta^*) \approx \epsilon_i \neq 0$), causing $\sum r_i \nabla^2 r_i(\theta)$ to inject indefinite curvature into H_L at the minimizer. Because `EGN` evaluates curvature via $J^T J \succeq 0$, it converges stably to the $\mathcal{O}(\sigma)$ noise floor ($860\times$ lower error at $\sigma = 10^{-3}$ and $460\times$ lower at $\sigma = 10^{-2}$).